

FROZEN SITE-PROBE ACCURACY IS NOT A STANDALONE HARMONIZATION SCORE: AN AUDIT OF EVALUATION PRACTICE FOR MULTI-SITE MRI

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ABSTRACT

Image-level MRI harmonization is often judged with a site probe, a classifier that predicts the acquisition site: lower accuracy on harmonized images is read as success. We examine how reliable such verdicts are, starting from a frozen probe trained on unharmonized images, on raw ABIDE I T1-weighted MRI (1,112 subjects, 17 sites) with nine harmonized outputs as test cases. Verdicts depended on probe training: five seeds of one recipe gave the same CycleGAN output balanced accuracies from 0.13 to 0.59, and five longer-trained, stable probes found the site more decodable than the frozen probe in eight of nine outputs. A Gaussian blur, moving intensities away from the target, scored lowest on the frozen probe, at the shuffled-label baseline. Probes trained on harmonized images decoded the source site far better than raw-trained probes (+0.42 to +0.55 balanced accuracy, paired intervals above zero), also on brain-only slices and their intensity histograms, so low frozen-probe accuracy does not establish removal of site-label information. The output with the smallest intensity-distribution change barely moved toward the target. We recommend several well-trained probes, including output-trained ones, against a shuffled-label baseline, with target alignment reported apart from image change.

Index Terms— MRI harmonization, evaluation, site classification, domain shift, reproducibility

1. INTRODUCTION

Acquisition differences between sites confound pooled MRI analyses and degrade models on unseen scanners [1, 2]. Image-level harmonization translates scans toward a common appearance. Travelling subjects [3] or paired protocols [4] allow direct checks on few scanners, so large retrospective cohorts are judged indirectly [5]. A common proxy is a *site probe*: a classifier that predicts the acquisition site from an image, which is easy on raw MRI [2]. If its accuracy drops on harmonized images, site effects are taken to be

removed [6, 7]; this is reported alongside similarity to the input, such as PSNR, and intensity-distribution distances. Some studies retrain the probe on harmonized images [6] or on image features [7, 8]. Others train it on original images and apply it unchanged to harmonized ones: the PRISM method [9], for example, reports this accuracy drop, with anatomy and segmentation checks, as evidence of harmonization. This frozen-probe protocol is our starting point.

Each proxy measures less than it appears to. A frozen probe reports what one classifier recovers; it can also fail because outputs are unfamiliar, damaged or noisy. Removing an attribute until a fixed adversary fails does not stop a new classifier from recovering it [10]. Site is also decodable at about 0.9 chance-corrected balanced accuracy from frozen, even randomly initialized, encoders, so it lies partly in low-level image statistics [11]. Similarity to the input rewards doing nothing, and small change says nothing about the target.

We examine how reliable site-probe verdicts are, starting from the frozen probe of an earlier benchmark, on outputs of six harmonization families. Site classifiers retrained after ComBat [12] or histogram matching [13] still predict site; we quantify how probe training changes the verdict and whether low frozen-probe accuracy survives probes trained on harmonized outputs, separating geometry from intensity, measuring target alignment and adding a blur control. We contribute a protocol with paired uncertainty and probe controls, these findings and a four-item reporting checklist (Sec. 5).¹ We neither propose nor rank harmonizers.

2. DATA AND AUDITED OUTPUTS

Cohort. Raw ABIDE I T1-weighted scans [14] of 1,112 subjects from 17 sites are split by subject, stratified by site, into 890 training, 113 validation and 109 test subjects. Volumes are reoriented (in-plane rotation still varies), clipped to $[p_1, p_{99}]$ and scaled to $[0, 1]$; one axial slice per subject is chosen by foreground area, cropped to a dilated head mask and resized to 256×256 . NYU, the largest site, is the target;

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¹Code, the evaluation protocol with all amendments, and per-subject results: <https://github.com/muqsitamir/HarmonIt>.

19 test subjects are from NYU and 90 from 16 source sites. Slices are frozen and verified for every output. The label is coarser than the acquisition: ABIDE I releases UM, UCLA and Leuven as two sub-samples each, merged by our labels, and 11 of 17 sites hold more than one voxel size or matrix (UCLA includes 10 scans at $1.5 \times 1.5 \times 4$ mm); no per-subject scanner identifier is released.

Audited outputs. Earlier benchmark outputs were evaluated without retraining: NeuroCombat [1, 15] per pixel, fit on the test cohort itself (transductive); histogram matching of foreground quantiles to a reference pooled over all 17 training sites, not NYU, as in the benchmark [16]; CycleGAN [17]; StarGAN [18], aggressive and conservative; a DLEST-style baseline without DLEST’s energy-based sampler [6], after 1,000 and 1,500 steps; conditional diffusion image-to-image translation [19, 20] (strength 0.35, 50 DDIM steps), sampled twice from one checkpoint. This gives nine outputs. As a control that ignores the target, raw test slices were blurred with a Gaussian ($\sigma = 2, 4$ and 8 pixels). Learned methods were trained on the training split. Test subjects were inspected in the earlier benchmark, so this is a retrospective reanalysis; each audit experiment was written into a protocol before its outcome was seen.

3. AUDIT PROTOCOL

Change and target alignment. PSNR on a fixed $[0, 1]$ scale and normalized cross-correlation (XCorr) compare output with input on whole images; target-conditioned generators return NYU inputs almost unchanged, so we summarize the 90 source subjects. *Change* is the per-subject Wasserstein-1 distance W between input and output intensities. *Target alignment* is the Wasserstein-1 distance and KL divergence between foreground intensities (defined on the raw slice) and a reference pooled from all 147 NYU training subjects; ΔW_{NYU} is the output’s distance minus the input’s.

Site probes. Image probes are single-channel ResNet-18s [21] predicting 17 sites, tested on the fixed slices; unless stated otherwise they use AdamW (learning rate 3×10^{-4} , batch 64, 500 steps) and random affine augmentation and are selected on validation BA. We use 53 probes in five families, 44 of them ResNets trained by us:

Frozen (1): the earlier benchmark’s own checkpoint.

Benchmark recipe (5 seeds): that training repeated, on a random axial slice of a raw training volume at each step. Does the seed change the verdict?

Converged (5 seeds): four times the steps, cosine decay and reseeded data-loader workers, tested at the final epoch. Do they agree?

Shuffled labels (4): labels permuted between subjects. Is the probe reading images, not labels?

Slice probes (24, plus 6 restricted-input networks and 8 logistic fits): trained on one fixed slice per training subject,

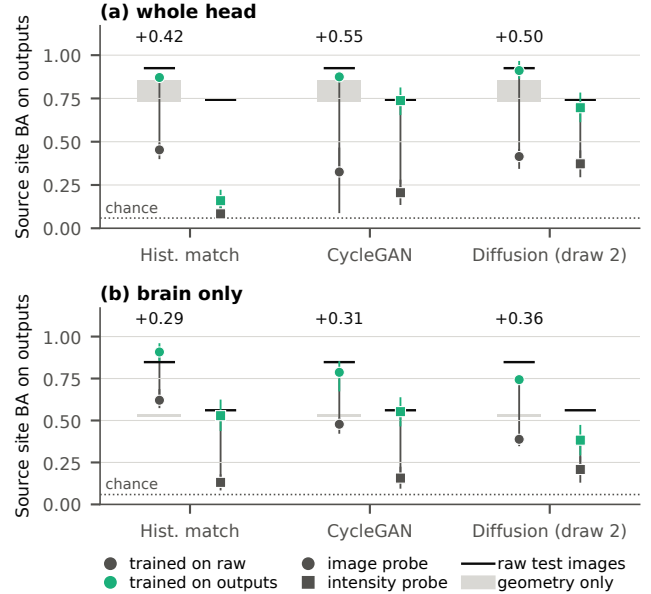


Fig. 1. Probes trained on raw slices (grey) or on a method’s outputs (aqua), tested on that method’s outputs (diffusion: second draw), on (a) whole heads and (b) brain-only slices: image probes (circles, mean and range over 3 seeds) and intensity-histogram probes (squares, 95% interval). Numbers: output- minus raw-trained image probes (paired, three-seed mean BA). Black bar: raw-trained probe on raw test slices. Grey: image probes given only head silhouettes or brain masks.

from raw images or one method’s outputs, selected on validation and tested on test subjects. Is site still decodable? Restricted variants see only head silhouettes or brain masks (geometry), only foreground intensity histograms (intensity; logistic regression), or slices inside raw-volume HD-BET brain masks [22].

The three slice-probe methods were chosen in advance to span intensity, adversarial and diffusion families; the converged, silhouette, histogram and brain-only probes were added after earlier results but before their own.

Statistics. Source BA averages the 16 source sites (2–15 test subjects each). A uniform guess over the 17 classes scores $1/17 \approx 0.06$ and one class predicted for every subject $1/16$; shuffled-label probes, our empirical null, scored 0.03–0.08. Intervals are 95% percentiles of 2,000 within-site subject bootstrap replicates, paired across methods and probes, and conditional on these sites and checkpoints.

4. RESULTS

Probe validity. Raw source BA was 0.95 for the frozen probe, 0.81–0.98 for benchmark-recipe and 1.00 for converged probes. Probes learn site-related structure, not only

Table 1. Source-subject means ($n = 90$). PSNR, XCorr, W : output vs. input; ΔW_{NYU} (negative is closer), KL_{NYU} : distance to NYU. Site BA: frozen probe with 95% interval and share labelled NYU; 5 benchmark-recipe and 5 converged probes: mean [min, max]; blur: frozen probe only. Shuffled labels: 0.03–0.08. [†]Fit on the test cohort. [‡]Pooled 17-site reference, not NYU.

Method	Change			Target alignment		Frozen probe		Probes retrained on raw images	
	PSNR	XCorr	W	ΔW_{NYU}	KL_{NYU}	Source BA	NYU	Benchmark recipe	Converged recipe
Raw input	–	–	0	0	0.30	0.95	0%	0.90 [0.81, 0.98]	1.00 [1.00, 1.00]
NeuroCombat [†]	20.0	0.940	0.055	+0.003	0.88	0.27 [0.21, 0.32]	2%	0.33 [0.26, 0.37]	0.34 [0.28, 0.42]
Histogram matching [‡]	23.0	0.982	0.054	-0.053	0.09	0.44 [0.37, 0.52]	10%	0.44 [0.27, 0.59]	0.61 [0.55, 0.68]
CycleGAN	21.3	0.977	0.048	-0.036	0.25	0.31 [0.25, 0.37]	52%	0.28 [0.13, 0.59]	0.49 [0.44, 0.54]
StarGAN (aggr.)	16.0	0.896	0.071	-0.019	0.32	0.21 [0.15, 0.27]	56%	0.35 [0.18, 0.54]	0.70 [0.65, 0.81]
StarGAN (cons.)	26.9	0.991	0.021	-0.021	0.27	0.83 [0.79, 0.86]	1%	0.78 [0.66, 0.89]	0.98 [0.94, 0.99]
DLEST-style 1500	26.6	0.993	0.024	-0.020	0.25	0.67 [0.61, 0.72]	10%	0.73 [0.65, 0.82]	0.96 [0.95, 0.97]
DLEST-style 1000	28.4	0.993	0.017	-0.009	0.31	0.71 [0.65, 0.77]	6%	0.84 [0.76, 0.89]	0.99 [0.96, 1.00]
Diffusion, draw 1	21.7	0.948	0.008	-0.001	0.83	0.37 [0.32, 0.42]	36%	0.42 [0.24, 0.63]	0.57 [0.52, 0.59]
Diffusion, draw 2	21.7	0.948	0.008	-0.002	0.82	0.35 [0.29, 0.40]	38%	0.42 [0.30, 0.62]	0.56 [0.50, 0.59]
Blur control, $\sigma=2$	24.7	0.974	0.018	+0.016	0.57	0.07 [0.06, 0.10]	0%	–	–

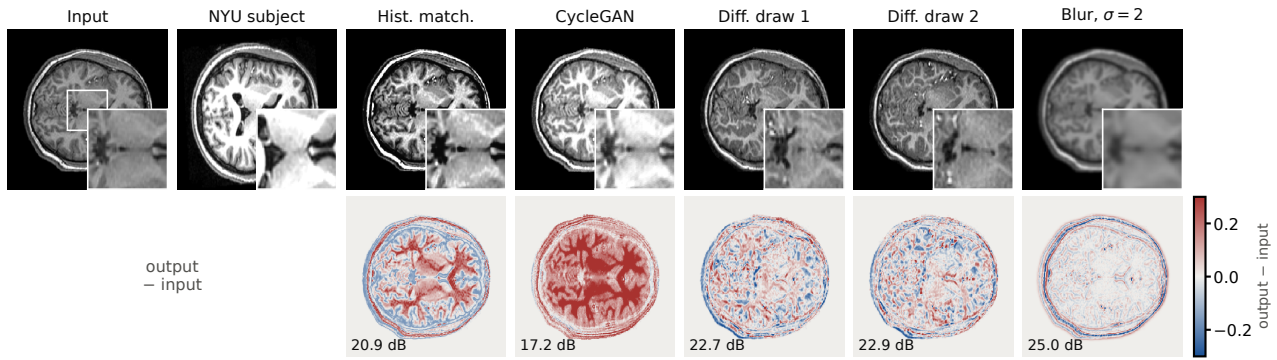


Fig. 2. Median-PSNR source subject and a matched axial NYU test subject, in the input’s display window; insets: $2\times$ zoom of the box. Bottom: output – input, with PSNR. Diffusion adds texture that differs between draws; blur has the highest PSNR.

scanner appearance: head silhouettes alone gave 0.73–0.86. Age, sex and diagnosis alone gave 0.22 [0.15, 0.29] (0.31 with brain volume).

Verdicts depend on probe training. Benchmark-recipe probes assigned very different site BA to the same outputs (Table 1): CycleGAN 0.13–0.59 and aggressive StarGAN 0.18–0.54, with Kendall τ 0.39–0.83 between method orderings. Their validation BA fluctuated between epochs, and raw accuracy did not reveal it: the probe with raw BA 0.98 gave CycleGAN 0.20. Much of this spread was a training-stability effect: converged probes (validation BA within 0.035 over the last five epochs) varied far less across seeds (τ 0.78–0.94; per-output ranges at most 0.16), but every converged probe exceeded the frozen probe’s interval on all outputs except NeuroCombat, e.g. aggressive StarGAN 0.65–0.81 against 0.21. Orderings changed (τ 0.56–0.67 with the frozen probe).

A blur scored lower than every harmonizer. A $\sigma = 2$ blur kept PSNR at 24.7 dB, above CycleGAN and diffusion, and moved intensities away from NYU ($\Delta W_{\text{NYU}} = +0.016$), yet scored frozen-probe BA 0.07 [0.06, 0.10], in the shuffled-label range ($\sigma = 4, 8$: 0.06, 0.07). The probe labelled 87 of 90 blurred slices Yale and none NYU; on CycleGAN, aggressive StarGAN and diffusion, 65–80% of its errors named NYU.

Unfamiliar images alone can defeat a frozen probe.

Small change is not target alignment. Diffusion changed the intensity distribution least ($W = 0.008$) yet barely moved it toward NYU ($\Delta W_{\text{NYU}} = -0.0013$ [–0.0023, –0.0004], 1% of the raw distance) and raised KL divergence to NYU from 0.30 to 0.83. Outputs are also a random draw: a second sample from the same checkpoint changed images more than harmonization did (mean absolute foreground difference, median 0.085 against 0.075), at unchanged PSNR.

Is site still decodable? Slice probes trained on raw slices reached BA 0.92 on raw test slices but 0.45, 0.33 and 0.41 on histogram-matched, CycleGAN and diffusion outputs, a drop interpreted as evidence of harmonization [9]. Probes trained on each method’s outputs recovered the source site from its test outputs with BA 0.87, 0.87 and 0.91 (Fig. 1a), exceeding raw-trained probes by +0.42 [0.36, 0.48], +0.55 [0.52, 0.58] and +0.50 [0.45, 0.54] (paired differences of unrounded three-seed mean BA). These probes use the shorter benchmark recipe (Table 1), so they may understate decodability. Because head geometry identifies sites, we repeated this on brain-only slices: raw-trained probes reached 0.85 on raw brains but 0.62, 0.48 and 0.39 on the outputs, and

probes trained on outputs 0.91, 0.79 and 0.74 (Fig. 1b; differences +0.29 to +0.36, all intervals above zero, also with final-epoch checkpoints); brain masks alone gave 0.52–0.54. On whole heads, an output-trained intensity-histogram probe recovered little from histogram matching (0.16 [0.11, 0.22]), whose head histograms are matched by construction, but recovered CycleGAN (0.74) and diffusion (0.70) as well as raw ones (0.74). Within the brain it recovered all three: 0.53 [0.44, 0.62], 0.55 [0.46, 0.64] and 0.38 [0.29, 0.47], against 0.56 on raw brains and 0.13–0.21 for the raw-trained probe. Low frozen-probe accuracy does not establish removal of site-label information, much of which output-trained probes decode.

5. DISCUSSION AND RECOMMENDATIONS

One site number is not a verdict: it moved by up to 0.46 across seeds and was lowest for a blur. We recommend reporting (1) several converged probes, their recipe, raw accuracy, a shuffled-label null and a fidelity measure; (2) probes trained on the outputs, with channel-restricted and geometry controls, whose high accuracy shows retained site-label information, not by itself failed harmonization, as head and brain shape predict site; (3) intensity alignment to the target apart from change, on source subjects: the output with the smallest change closed 1% of its distance to NYU; and (4) paired uncertainty over subjects and generator draws.

Limitations. We use one cohort, one target site and one axial slice per subject, so volumetric effects are not assessed; with 2–15 test subjects per source site, intervals are wide. Without anatomical ground truth, PSNR and XCorr measure agreement with the input, not correctness. Brain masks leave brain shape, slice level and orientation, and site is confounded with demographics and with acquisition geometry that varies within a site; our controls bound these channels but cannot separate scanner effects from biological differences. Generators were not retuned, and one baseline is transductive.

6. COMPLIANCE WITH ETHICAL STANDARDS

This research study was conducted retrospectively using human subject data made available in open access by ABIDE I [14], whose sites shared the fully anonymized data with local ethics approval or an explicit waiver. Ethical approval was not required as confirmed by the license attached with the open access data.

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